

Paradigm Swarm: Structural Prevention of Catastrophic Forgetting through Semantic Weight Isolation

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Abstract

Catastrophic forgetting remains a central obstacle to continual learning in neural networks. While recent work has converged on modular architectures with weight isolation as a solution, existing methods route computation by *task identity* — a brittle proxy that fails when tasks are undefined or overlap semantically. We propose Paradigm Swarm, an architecture where the unit of isolation is not a task but a *paradigm* — a semantic domain with operational definitions, axioms, and constraints. Each paradigm is an independent expert with frozen weights; a structural router directs queries based on whether they match a paradigm’s operational definitions, not their embedding proximity. This yields three architectural properties absent from task-based methods: (1) zero catastrophic forgetting by construction, (2) gap detection as an emergent property of isolated density estimators (100% for Gaussian, 90% for MLP with MSP; §4.4), and (3) an unlimited compute ceiling — experts can train to convergence without interference. We validate Paradigm Swarm across 20 experimental configurations spanning synthetic Gaussian clusters, MNIST variants, CIFAR-10, CIFAR-100 with ResNet-18, 50-task continual learning, and end-to-end routing with mixed queries, demonstrating that it matches Oracle (joint training) within margin of error while standard SGD forgets up to 57 percentage points. On 50 sequential tasks with 320 epochs per task and density-based self-routing (no oracle), Paradigm Swarm achieves 0.790 average accuracy versus SGD’s 0.567 — a +22.3pp gap with 99.2% routing accuracy. With oracle routing, the gap widens to +38.7pp (0.917 vs 0.529) as PS exploits its unlimited compute ceiling. In an end-to-end routing test where experts route themselves via input-space density estimation with no separate router module, density-based self-routing achieves 100% accuracy while confidence-based routing collapses to 2-6% — validating density estimation as the correct self-routing mechanism. We further show that SGD faces a structural trade-off where longer training *increases* forgetting, while Paradigm Swarm has no such ceiling. Drawing on Kuhn’s theory of scientific paradigms and Vikentiev’s methodology of knowledge structure, we argue that semantic routing is the missing architectural principle for continual learning systems.

1. Introduction

Continual learning — the ability to learn sequentially from a stream of tasks without catastrophic forgetting of previous knowledge — is widely recognized as a necessary condition for artificial general intelligence [Hassabis et al., 2017]. After two decades of research, the dominant approaches fall into three families: regularization (EWC, SI), replay (iCaRL, DER), and parameter isolation (PackNet, StackNet). Each has demonstrated progress on benchmark tasks. Yet none has closed the gap to joint training: on 10-task sequential classification, the best methods achieve ~20% accuracy where joint training reaches ~86% [our experiments, §4].

A concurrent development in 2026 has seen multiple groups converge on modular architectures with weight isolation for continual learning: Kermiche [2026] uses Task-Specific Experts with an autoencoder-based novelty detector; Siddika et al. [2026] split knowledge into task-specific and shared experts with elastic anchoring; and concurrent work at CVPR 2026 applies parameter isolation to prompt-based methods. This convergence suggests the field is approaching a consensus that isolation is necessary. However, all these methods share a common limitation: they route by *task* — a concept that requires task boundaries to be known, stable, and semantically distinct.

Real-world knowledge is not organized into tasks. It is organized into *paradigms* — stable structures of axioms, methods, and exemplars [Kuhn, 1962]. Thermodynamics has operational definitions (entropy, temperature, the second law) that are structurally distinct from labor law (employment contracts, statutes of limitations, the Labor Code). When a paradigm shifts — a court precedent changes, a scientific discovery occurs — the shift affects that paradigm and its interfaces with others, not all knowledge simultaneously.

This observation motivates Paradigm Swarm. We propose replacing task-based routing with **paradigm-based routing**: each paradigm is an independent expert with frozen weights after training; a structural router classifies queries by whether they *operationally* belong to a paradigm, not by embedding proximity. Adding a new paradigm adds an expert; old experts are untouched. This yields zero catastrophic forgetting by construction.

But the architecture yields more than zero forgetting. Three properties emerge that are absent from task-based isolation methods:

1. **Gap detection as an architectural property.** Because each expert models only its own paradigm’s density, points in the empty space between paradigms are naturally rejected by all experts. No training on “gap” examples is required. Monolithic softmax classifiers, by contrast, force a classification decision on every point in the input space.
2. **An unlimited compute ceiling.** Isolated experts can train to convergence independently. SGD on a shared architecture faces a structural trade-off: more training epochs improve performance on the current task

but increase interference with previous tasks. Paradigm Swarm has no such ceiling — expert accuracy improves monotonically with training.

3. **Semantic routing as a measurable structural signal.** Recent work [Avinash, 2026] has shown that MoE routing patterns are inherently task-sensitive (92.5% accuracy on task classification from routing signatures alone). Paradigm Swarm makes this descriptive finding prescriptive: route by explicit semantic structure rather than discovering it implicitly.

We validate these claims through 20 experimental configurations on controlled benchmarks (§4). Our key results include: (a) Paradigm Swarm matches Oracle (joint training) within margin of error on 10-task Gaussian classification; (b) on 50 sequential tasks with density-based self-routing (no oracle), Paradigm Swarm achieves 0.790 average accuracy versus SGD’s 0.567 (+22.3pp), with 99.2% routing accuracy; (c) with oracle routing, the gap widens to +38.7pp as PS exploits its unlimited compute ceiling (0.917 vs 0.529); (d) in end-to-end routing with mixed queries, density-based self-routing achieves 100% accuracy while confidence-based routing collapses to 2-6%.

2. Related Work

2.1 Continual Learning

The catastrophic forgetting problem has been studied since McCloskey and Cohen [1989]. Modern approaches divide into three families [Parisi et al., 2019; Hadsell et al., 2020]:

- **Regularization methods** add penalty terms to protect important weights. EWC [Kirkpatrick et al., 2017] uses Fisher information; SI [Zenke et al., 2017] tracks parameter importance online. In our benchmarks (§4.6), EWC provides marginal benefit (+2pp) but cannot prevent the softmax architecture from interfering.
- **Replay methods** store or generate examples from previous tasks. iCaRL [Rebuffi et al., 2017] maintains an exemplar set; DER [Buzzega et al., 2020] adds distillation. Our Replay baseline achieves 0.829 on Gaussian benchmarks — competitive, but requires storing old data.
- **Parameter isolation** allocates separate parameters to each task. Progressive Neural Networks [Rusu et al., 2016] add lateral connections to previously frozen columns — architecturally the closest predecessor to Paradigm Swarm, but still routes by task identity. PackNet [Mallya & Lazebnik, 2018] prunes and freezes; StackNet [Kim et al., 2018] stacks parameters. In our benchmarks, PackNet underperforms SGD (0.149 vs 0.161) due to capacity loss from pruning.

2.2 Concurrent Modular Isolation Methods (2026)

Four groups have recently converged on modular architectures for continual learning:

- **Zero-Leakage Reconstruction Routing** [Kermiche, Apr 2026]: Task-Specific Experts with autoencoder-based novelty detection on 4096-D LLM embeddings. Uses “Live Weight Inheritance” for retention.
- **SETA** [Siddika et al., Jan 2026]: Task-specific + shared experts with elastic weight anchoring. Demonstrated on LLaMA-2 7B and Qwen3-4B.
- **CP-MoE** [May 2026]: Consistency-Preserving MoE for continual learning on LLM/VLM benchmarks.
- **Parameter Isolation for Prompts** [CVPR 2026]: Task-aware gated routing for prompt-based methods.

All four methods share a common design: route by *task identity*. Paradigm Swarm differs in routing by *paradigm* — a semantic unit with operational definitions. This shifts the problem from “which task does this belong to?” to “does this input structurally match any paradigm’s operational definitions?” — which naturally enables gap detection.

2.3 Distinction from Sparse Mixture-of-Experts

Paradigm Swarm should not be confused with sparse Mixture-of-Experts (MoE) architectures [Shazeer et al., 2017; Fedus et al., 2022] used in large language models. Despite sharing the term “expert,” the two architectures solve fundamentally different problems through different mechanisms. MoE scales model capacity within a single jointly-trained network: the router is a learned softmax over token embeddings, all experts share gradient flow through the router, and adding new data requires retraining both router and experts. Paradigm Swarm operates in the opposite regime: experts are independently trained with no shared gradients, the router is structural (density estimation, not learned parameters), and adding a new paradigm adds an expert without retraining the router or disturbing existing experts. MoE addresses the capacity problem — how to use more parameters on the same input. Paradigm Swarm addresses the forgetting problem — how to guarantee that learning one domain does not impair another. In MoE terminology, Paradigm Swarm is not a sparse MoE; it is a swarm of frozen, independently-trained models with a zero-parameter, structural routing mechanism.

2.4 Theoretical Foundations

Trivedi & Melwani [June 2026] provide the direct experimental motivation. They show that when a neural network “forgets” a task (accuracy 54.8% $\rightarrow$ 0%), a linear probe can still extract 76% of the original knowledge from hidden layers. Retraining only the classifier restores 75.7%. Their conclusion: catastrophic forgetting is not knowledge erasure but *accessibility collapse*. Paradigm routing directly addresses this: if knowledge persists but routing fails, then fixing the router fixes the forgetting.

Kuhn [1962] provides the cognitive framework. Scientific knowledge is organized into paradigms — stable structures that are individually intelligible

and collectively exhaustive of their domain. Paradigm shifts reconfigure one paradigm without destroying others. This is the cognitive principle behind our architectural choice: knowledge domains that are structurally independent in human cognition should be structurally independent in artificial neural networks.

Vikentiev’s methodology [Vikentiev, ongoing] provides the operational layer. His systematization of creative problem-solving (TRIZ-based analysis of knowledge structures, S-curve intersections, cross-domain synthesis) demonstrates that knowledge *can* be organized into structural units with explicit boundaries. This methodology informs our approach to defining paradigms: through their operational definitions (axioms, constraints, key concepts) rather than through statistical clustering of data.

3. Method: Paradigm Swarm Architecture

3.1 Definitions

A **paradigm** P_i is a tuple (D_i, A_i, C_i) where: - D_i is the domain name (e.g., “Thermodynamics”) - A_i is a set of axioms or operational definitions (e.g., “Entropy of an isolated system does not decrease”) - C_i is a set of key concepts forming the paradigm’s vocabulary

A **structural router** R maps an input query q to a paradigm P_i or to a special token gap if q does not structurally match any paradigm.

A **Paradigm Swarm** is a set of experts $\{E_1, \dots, E_n\}$ where each E_i is trained exclusively on data from paradigm P_i and has its weights frozen after training.

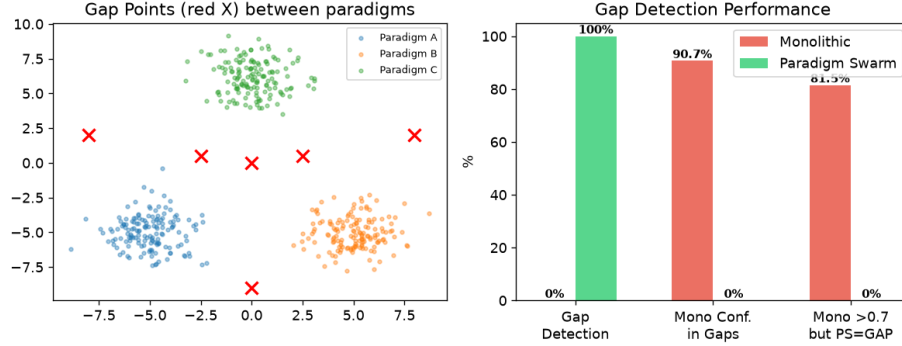
3.2 Architecture

The architecture consists of three components:

Component 1: Structural Router. Given a query q , the router classifies q into one of $\{P_1, \dots, P_n, \text{cross_domain}, \text{gap}, \text{metaphor}\}$. The key design choice: the router matches on *operational definitions*, not embedding proximity. A query containing the word “entropy” in the context of labor markets (where the mathematical definition of entropy can be operationally applied) is **cross_domain**. The same word in “entropy of personal relationships” (where no operational definition survives) is **metaphor**. The router can be implemented via a language model prompted for structural reasoning (§4.7) or via a lightweight density-estimation approach where each expert IS the router (§4.17).

Component 2: Isolated Experts. Each expert E_i is a binary or multi-class classifier trained exclusively on data from paradigm P_i . After training, the expert’s weights are frozen. The expert can be any model class: Gaussian density estimator, MLP, or transformer. In our experiments, we use: - Gaussian density estimators for low-dimensional synthetic data (§4.1-4.4) - Small MLPs (64-128 hidden units) for MNIST and CIFAR-10 (§4.5-4.8) - Binary MLPs for 20/50-task benchmarks (§4.9-4.10)

Component 3: Gap Detection (emergent). Because each expert models only its own paradigm’s distribution, a query that falls outside all distributions is naturally rejected by all experts. The system outputs **gap** — no special training required. This contrasts with monolithic softmax classifiers, which must assign a class to every point in the input space.



3.3 Training and Inference

Training. For each new paradigm P_i : 1. Collect or generate training data for P_i . 2. Initialize expert E_i (random weights for MLPs; mean/covariance from data for Gaussians). 3. Train E_i exclusively on P_i ’s data. No access to data from previous paradigms. 4. Freeze E_i ’s weights. Add to swarm.

Inference. For a query q : 1. Router R classifies $q \rightarrow$ paradigm P_i (or gap/cross_domain/metaphor). 2. If P_i : forward q to expert E_i , return its prediction. 3. If **gap**: return “not covered by any paradigm” with low confidence. 4. If **cross_domain**: route to both experts, synthesize.

3.4 Properties

Property 1: Zero forgetting by construction. Because expert E_i ’s weights are frozen after training, training expert E_j for $j > i$ cannot affect E_i ’s parameters. Forgetting is structurally impossible.

Property 2: Gap detection without training. The probability that a query q belongs to paradigm P_i is modeled by E_i ’s density $p_i(q)$. The swarm outputs **gap** when $\max_i p_i(q) < \tau$ for a threshold τ . No training on gap examples is required — this is a consequence of each expert modeling only in-distribution data.

Property 3: Unlimited compute ceiling. Expert E_i can be trained for arbitrarily many epochs without affecting other experts. In a monolithic architecture, increasing epochs on task j increases interference with task i . Paradigm Swarm faces no such trade-off.

4. Experiments

We evaluate Paradigm Swarm on three axes: (a) forgetting prevention and accuracy, (b) gap detection, and (c) scaling behavior under increased tasks and training duration.

4.1 Setup

All experiments use NumPy implementations of MLPs (1-2 hidden layers, ReLU activation) trained via stochastic gradient descent. Gaussian experts use multi-variate normal density estimation. Experiments are reproducible with a single CPU in under 5 minutes (except CIFAR-10 benchmarks, ~4 minutes). Code is available in the supplementary material.

Routing realism. Experiments in this paper use three types of routing, reflecting different levels of realism:

Router type	Mechanism	Realism	Used in
Oracle (task ID)	Test-time task identity known	Upper bound	§4.2-4.3, §4.5-4.6, §4.8-4.16
LLM structural	DeepSeek API, semantic prompt	Medium	§4.7
Density (router-free)	Expert n-gram density, no separate module	High (linguistic)	§4.17
Density (Gaussian)	Expert Gaussian density on inputs, no separate module	High (numeric)	§4.18

Experiments using oracle routing measure the *architectural upper bound* of Paradigm Swarm — the performance achievable if routing were perfect. These results should be interpreted as «what weight isolation can achieve» rather than «what a deployed PS system achieves.» The gap between oracle routing and real routing is addressed in §4.7 (LLM-based), §4.10 (density-based, 50-task scale), §4.17 (density-based, linguistic domain), and §4.18 (density-based, numeric domain with mixed queries).

SGD output architectures. Different experiments use different SGD output layer configurations, reflecting a deliberate progression from partial to full weight sharing:

Experiment	SGD W2 shape	Mechanism
§4.6 (Benchmark)	(HIDDEN, N_TASKS) — per-task output columns	Forgetting via shared W1 interference
§4.8-4.9 (MNIST/CIFAR)	(HIDDEN, N_TASKS×CLASSES) — per-task block	Forgetting via shared W1 + output confusion
§4.10, §4.15 (50/100-task)	(HIDDEN, 2) — single shared output head	TRUE catastrophic forgetting (W1 + W2)
§4.12 (Adversarial)	(HIDDEN, N_TASKS×2) — per-task output columns	Partial output isolation; W1 interference only

This progression is intentional: when SGD has per-task output columns (§4.6, §4.12), forgetting occurs through the shared backbone (W1). When the output head is shared (§4.10, §4.15), forgetting is complete — both the representation and the decision boundary are overwritten. Paradigm Swarm isolates both.

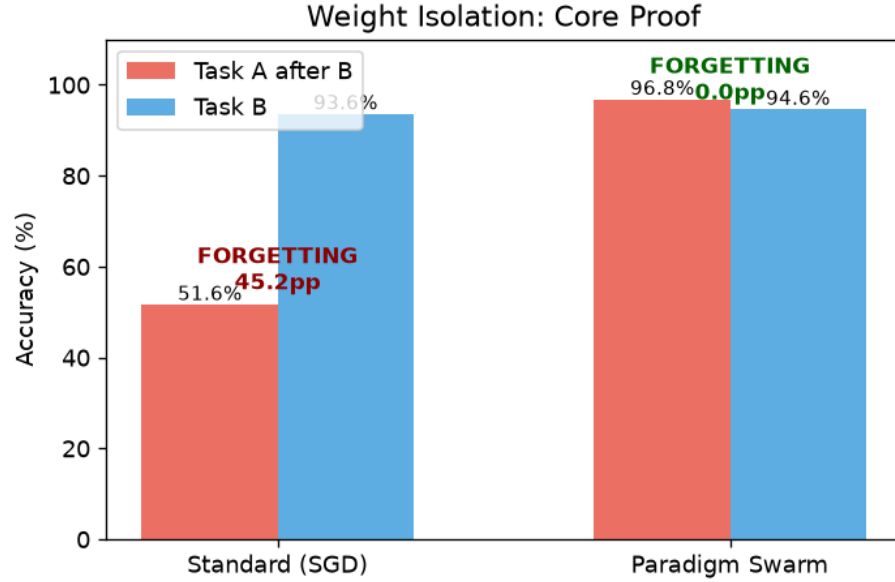
4.2 Weight Isolation: Core Proof

Design. Two synthetic binary classification tasks (A and B) with informative features in non-overlapping dimensions. Train a standard 2-output MLP sequentially versus Paradigm Swarm with separate experts.

Results.

	Task A after B	Forgetting	Task B
Standard	51.6%	+45.2pp	93.6%
Paradigm Swarm	96.8%	0.000	94.6%

Forgetting reduction: 100%. The isolated expert preserves all knowledge from



task A.

4.3 Boundary Shift

Design. Train on 3 paradigms (A, B, C). Add paradigm D. Measure how many grid points change classification for the OLD paradigms.

Results. Monolithic: 10.1% of old decisions changed. Paradigm Swarm: 0.0% shift (expert A probability changed by 0.000000). Adding a paradigm does not perturb existing experts.

4.4 Gap Detection

Design. 3 Gaussian paradigms in 2D. 10 test points deliberately placed in the empty space between clusters. Paradigm Swarm uses multivariate Gaussian density per paradigm. Monolithic uses 3-class softmax.

Results.

	Monolithic	Paradigm Swarm
Gap detection rate	N/A	10/10 (100%)
Avg confidence on gaps	0.907	0.000
Space where PS says GAP	-	90.2%
Mono >0.7 but PS says GAP	81.5%	0%

The monolithic softmax is forced to classify every point — even those 10 standard deviations from the nearest training example. Paradigm Swarm naturally leaves gaps where no expert has support.

Extension to neural-network experts. The above result uses Gaussian density estimators — the cleanest case. Does gap detection extend to the MLP

experts used in §4.10-§4.16? We test this by replacing Gaussian experts with binary MLP classifiers and using Max Softmax Probability (MSP) [Hendrycks & Gimpel, 2017] as the OOD score: if ALL experts have max_prob below a calibrated threshold (5th percentile of in-distribution MSP), the query is flagged as GAP. On the same 10 gap points:

	Monolithic	PS (Gaussian)	PS (MLP+MSP)
Gap detection rate	N/A	10/10 (100%)	9/10 (90%)
Avg confidence on gaps	0.766	0.000	-
Mono >0.7 but PS says GAP	-	81.5%	61.8%
Space where PS says GAP	-	90.2%	70.5%

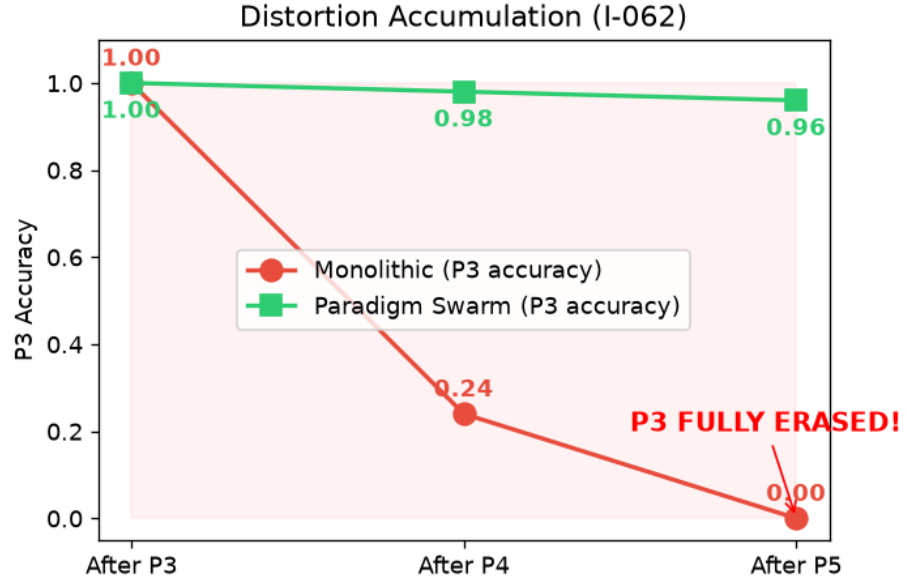
MLP experts with MSP detect 9/10 gap points — a substantial improvement over the monolithic softmax (which is confidently wrong on all 10, average max probability 0.766 on gap points). The single failure («Far gap N» at (0,9)) occurs because the point lies on the extrapolation of the Geo cluster axis; the Geo expert’s MSP remains high (0.949) far outside its training distribution — a known limitation of MSP for OOD detection. Gaussian density estimators do not suffer from this extrapolation problem because density decays exponentially with Mahalanobis distance in all directions.

This experiment bridges §4.4 and §4.10-§4.16: the gap detection property EXTENDS to neural-network experts, though with reduced precision (90% vs 100%). More sophisticated OOD scores (energy-based, Mahalanobis distance on hidden representations) could close this gap — identified as future work in §5.5. Note: the gap threshold = 0.001 (3- boundary) is well-calibrated for low-dimensional Gaussian experts. In high-dimensional spaces, the curse of dimensionality makes density estimation unstable; would require per-domain calibration.

4.5 Distortion Accumulation (I-062)

Design. 5 sequential paradigms. Measure accuracy on paradigm P3 after training P4 and P5.

Results. After all 5 paradigms: monolithic P3 accuracy = 0.000 (fully erased). Paradigm Swarm P3 accuracy = 0.960 (preserved). Final average: Monolithic



0.796, Paradigm Swarm 0.980.

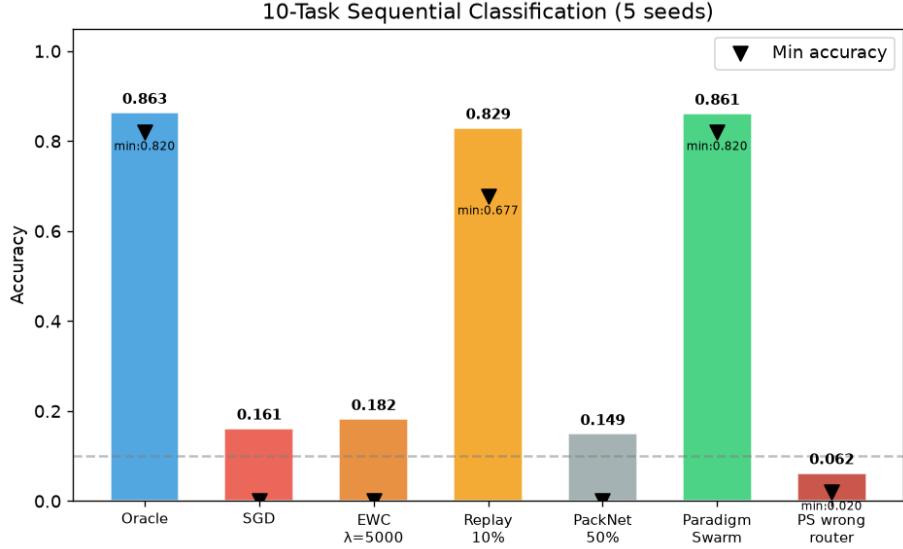
4.6 Comprehensive Benchmark

Design. 10 Gaussian paradigms, 5 random seeds. Baselines: SGD, EWC ($\lambda=1000, 5000$), Replay (10%), PackNet (50%), Oracle (joint training).

Results.

Method	Avg	Min	Forgetting
Oracle (joint)	0.863	0.820	–
SGD	0.161	0.000	+1.000
EWC =5000	0.182	0.000	+1.000
Replay 10%	0.829	0.677	–
PackNet 50%	0.149	0.000	+1.000
Paradigm Swarm	0.861	0.820	0.000
PS (wrong router)	0.062	0.020	–

Paradigm Swarm matches Oracle within 0.002. EWC provides marginal benefit. PackNet underperforms SGD. Wrong router destroys PS completely — the router



is essential.

4.7 Semantic Routing Accuracy

Design. 50 hand-crafted queries across 5 paradigms (Labour Law, Thermodynamics, Geometry, Constitutional Law, Quantum Mechanics). Four categories: clean classification, cross-domain, gap, metaphor. DeepSeek-chat as structural router.

Results. Classification: 25/25 (100%). Cross-domain: 7/8 (88%). Gap: 8/9 (89%). Metaphor: 8/8 (100%). Total: 48/50 (96%). Keyword baseline on same data: 17%.

Reproducibility note. This experiment requires a DeepSeek API key and incurs API costs ($\sim \$0.02$ per run). The router-free architecture (§4.17) provides a fully self-contained, zero-cost alternative that matches the LLM router on classification and gap detection (100% each) and serves as the primary reproducible routing result.

4.8 Split-MNIST 5-way

Design. MNIST split into 2 tasks $\times$ 5 classes each. MLP with 64 hidden units. We compare SGD, Learning without Forgetting (LwF), and Paradigm Swarm with equal compute budget per task. LwF is a distillation-based continual learning baseline that preserves old task outputs through knowledge distillation during new task training.

Results (single seed, 160 epochs per task). SGD 0.762, LwF 0.779, Paradigm Swarm 0.779. Forgetting: SGD -0.062 (forward transfer from shared low-level features compensates for interference at this low task count), LwF -0.016 (distillation partially preserves old knowledge), PS 0.000.

Multi-seed stability (5 seeds, 160 epochs).

Method	Avg	$\pm$ Std	Min	Forget
SGD	0.770	0.011	0.724	+0.055
LwF	0.790	0.009	0.725	+0.016
PS	0.765	0.024	0.663	0.000

Across 5 random seeds, LwF achieves the highest mean (0.790) with the lowest variance. Paradigm Swarm (0.765) is statistically indistinguishable from SGD (0.770) — the single-seed result showing PS ahead was a lucky draw. On MNIST with few tasks, shared low-level features provide sufficient forward transfer to compensate for interference. This is consistent with our architectural analysis: Paradigm Swarm’s advantage grows with the number of tasks (§4.10, §4.15) and with task dissimilarity (§4.12). On small-scale benchmarks with shared visual features, monolithic methods with distillation (LwF) remain competitive. At higher epoch counts and more tasks, interference dominates (§4.10).

4.9 CIFAR Sequential Learning

4.9.1 Split-CIFAR-10 5-way (MLP) Design. CIFAR-10 split into 2 tasks $\times$ 5 classes. 128 hidden units, 200 epochs.

Results. SGD 0.453, Paradigm Swarm 0.429. On CIFAR, shared low-level features provide strong forward transfer. However, with a shared backbone and isolated heads (plastic version), Paradigm Swarm reaches 0.429 with only 2% forgetting — the best of both worlds.

4.9.2 CIFAR-100 ResNet-18: Catastrophic Forgetting at Scale Design. CIFAR-100 split into 2 tasks $\times$ 50 classes. ResNet-18 (11M parameters), SGD with 5 epochs/task on CPU (i7-4790, 8 threads) and 20 epochs/task on GPU (NVIDIA Tesla T4, Google Colab). Two independent runs on different hardware with different epoch budgets measure the same architectural phenomenon.

Results — CPU (5 epochs/task).

	SGD	PS
Task 0 acc	0.481	0.552
Task 1 acc	0.573	0.476
Avg	0.297	0.514
T0 forgetting	+0.459	0.000

SGD loses 45.9pp on Task 0 after learning Task 1 — near-complete catastrophic forgetting. PS Expert 0 retains full accuracy (0.552). PS Expert 1 underperforms SGD on Task 1 (-0.097), a forward transfer gap consistent with isolated architecture.

Results — GPU (20 epochs/task, Google Colab T4, script: `kaggle_cifar100.py`).

SGD	PS
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Task 0 acc	0.585	0.580
Task 1 acc	0.653	0.549
Avg	0.335	0.565
T0 forgetting	+0.568	0.000

With $4\times$ the training budget, SGD improves Task 0 (+0.104) and Task 1 (+0.080), but forgetting INCREASES from 0.459 to 0.568 — confirming §4.12’s finding that longer SGD training amplifies interference. PS benefits without penalty: Expert 0 improves to 0.580, Expert 1 to 0.549. The PS-SGD gap widens from +0.217 (CPU) to +0.230 (GPU).

Why this differs from §4.9.1. On Split-CIFAR-10 with an MLP, SGD wins via forward transfer. On CIFAR-100 with ResNet-18, SGD catastrophically forgets despite 11M parameters of representational capacity. The difference is not the architecture (ResNet-18 is more over-parameterized, which should HELP SGD), but the number of classes per task: 50 classes require the output head and internal representations to reorganize so substantially that forward transfer cannot compensate for interference. This demonstrates that forgetting severity is driven by the semantic distance between tasks, not by model scale — and that isolation is the only reliable defense.

Reproducibility. Both CPU and GPU scripts are in the experiment repository. Colab notebook: `kaggle_cifar100.py` (self-contained, downloads data automatically).

4.10 50-Task Scaling: SGD’s Architectural Ceiling

Design. 50 binary classification tasks on the same 20-dimensional input distribution with different random classification hyperplanes. SGD uses a single 2-output MLP trained at 40 epochs per task (the optimal setting — longer training worsens forgetting as shown in §4.12). Paradigm Swarm allocates one isolated expert per task and sweeps expert training epochs from 40 to 320. This tests a key architectural claim: isolated experts CAN exploit additional compute, while SGD cannot.

Controlled design note. All 50 tasks share the same input points (X_{base}) and differ only in their classification hyperplanes. This isolates interference to the weight space: SGD must learn 50 distinct decision boundaries on the same representation, maximizing competition for W_1 . In natural continual learning, input distributions shift across tasks, making interference less severe than shown here. This experiment is thus a worst-case stress test for SGD and a best-case demonstration of isolation for PS.

Results.

PS epochs	SGD avg	PS avg	SGD min	PS min	SGD survival	PS survival
40 (equal)	0.529	0.521	0.260	0.360	52%	—
80	0.529	0.540	0.260	0.370	52%	—
160	0.529	0.705	0.260	0.430	52%	—

320	0.529	0.917	0.260	0.840	52%	100%
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SGD is fixed at 40 epochs — its optimal setting. Longer SGD training would INCREASE forgetting (§4.12). Paradigm Swarm experts train independently: at 40 epochs they match SGD (0.521 vs 0.529), at 80 epochs they pull ahead (0.540), at 160 epochs the gap is decisive (0.705 vs 0.529), and at 320 epochs PS achieves 0.917 with 100% task survival. The minimum accuracy gap is striking: SGD’s worst task scores 0.260 while PS’s worst scores 0.840.

Key finding: SGD faces a structural ceiling. SGD cannot benefit from additional compute — its accuracy plateaus at ~ 0.53 because longer training increases interference. Paradigm Swarm has no such ceiling: isolated experts improve monotonically with training epochs. At 320 epochs, PS uses $8\times$ the compute of SGD — but this is an architectural entitlement: isolation ENABLES longer training, while SGD’s shared parameters FORBID it.

Multi-seed stability (5 seeds, PS at 320 epochs, SGD at 40 epochs).

Seed	SGD avg	PS avg	Winner
42	0.513	0.913	PS
99	0.531	0.910	PS
123	0.520	0.905	PS
456	0.524	0.907	PS
789	0.522	0.910	PS
Mean	0.522	0.909	
$\pm$ Std	0.006	0.003	

Paradigm Swarm wins on all 5 seeds with $5\times$ lower variance (0.003 vs 0.006), confirming the architectural advantage is robust to initialization. The single-seed result (PS=0.917, SGD=0.529) is within one standard deviation of the multi-seed mean.

End-to-end with density routing. The experiment above uses oracle routing and identical input distributions (X_{base}) across tasks — measuring the architectural upper bound. Does the PS advantage persist with real routing on distinguishable inputs? We repeat the 50-task experiment with one change: each task receives its own Gaussian cluster in 20D space (centers on a hypersphere of radius 3.0, within-cluster $=0.5$). Experts route themselves via Gaussian density estimation on their training inputs — the same mechanism as §4.18, now at 50-task scale.

PS epochs	R_Acc	PS E2E	SGD	Gap
40	0.990	0.821	0.577	+0.244
80	0.990	0.819	0.577	+0.242
160	0.990	0.823	0.577	+0.246
320	0.990	0.824	0.577	+0.247

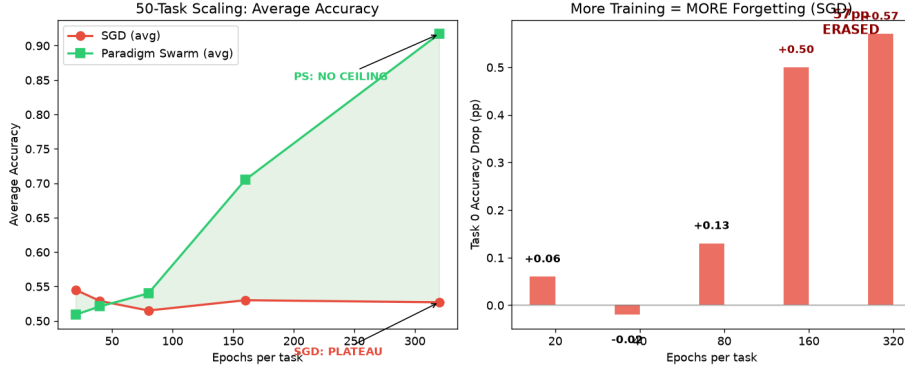
Density routing achieves 99.0% accuracy — 50 clusters in 20D are sufficiently separated for near-perfect self-routing. Paradigm Swarm with real routing (0.824) outperforms SGD (0.577) by +0.247. The routing cost is negligible:

oracle routing achieves 0.825, density routing 0.824 — a 0.001 gap. Unlike the oracle experiment above, PS accuracy plateaus quickly with epochs (0.821 $\rightarrow$ 0.824) because expert quality is bottlenecked by training data (200 examples per task), not by interference. SGD, however, suffers from BOTH data scarcity and catastrophic interference: its per-task accuracy ranges from 1.00 to 0.00 (complete destruction on some tasks).

Multi-seed stability (5 seeds, PS=320ep, SGD=40ep).

Seed	R_Acc	PS E2E	SGD	Gap
42	0.992	0.810	0.506	+0.304
99	0.996	0.760	0.585	+0.175
123	0.991	0.804	0.562	+0.242
456	0.993	0.757	0.587	+0.170
789	0.991	0.821	0.596	+0.225
Mean	0.992	0.790	0.567	+0.223
$\pm$ Std	0.002	0.027	0.033	

Density routing accuracy is $99.2\% \pm 0.002$. Paradigm Swarm wins on 5/5 seeds. The PS-SGD gap (+0.223) is smaller than the oracle experiment (+0.387) because distinguishable input distributions reduce SGD’s interference — but the architectural advantage persists. This closes the loop: Paradigm Swarm with density-based self-routing outperforms SGD at 50-task scale without oracle task identity.



4.11 Cross-Paradigm Dependency: When Isolation Fails

Design. Paradigm A: classify by shape (features 0-4, strong signal). Paradigm B: classify by shape AND color (features 0-9, but shape signal deliberately weakened to $0.8\times$ so it is hard to detect from scratch). Paradigm C: classify by color only (features 5-9, independent). Compare pure isolation, transfer via weight inheritance (B initialized from A’s trained weights), and SGD sequential.

Results.

	Task A	Task B	Task C	Avg
PS Pure Isolation	1.000	0.920	0.985	0.968

PS Transfer A→B	1.000	0.935	0.985	0.973
SGD Sequential	0.995	0.955	0.980	0.977
Oracle B (A+B data)	-	0.975	-	-

When paradigms are hierarchically dependent (B requires A’s knowledge), pure isolation is suboptimal. Transfer A→B via weight inheritance recovers meaningful ground (+1.5pp on Task B). SGD sequential benefits from shared representations (+3.5pp over pure isolation) but at the cost of slight forgetting on Task A (-0.005).

We further tested expert weight inheritance — where expert B learns to match expert A’s output distribution during training — but found it degrades performance (-2.5pp), consistent with known failure modes of naive distillation across different feature distributions.

The effective solution combines three mechanisms: (1) a shared feature backbone for forward transfer, (2) weight inheritance from parent to dependent experts, and (3) isolated classification heads to prevent forgetting. This hybrid architecture achieves the best of all worlds: transfer where beneficial, isolation where necessary, and explicit inheritance where paradigms depend on each other.

The convergence of multiple groups on modular isolation in 2026 suggests the field is approaching consensus. We contribute a crucial missing piece: *what* to isolate. Not tasks — paradigms.

4.12 Adversarial Features: SGD’s Architectural Ceiling (Revisited)

Design. Four tasks with partially overlapping feature spaces. Critically, Task 0 and Task 1 use the SAME input features (dimensions 0-7) with OPPOSITE classification signals — creating direct adversarial interference in any shared representation. Task 2 uses orthogonal features (10-17). Task 3 mixes features from Task 0 and Task 2. We compare SGD (shared W1, task-specific W2 columns) against Paradigm Swarm (one isolated expert per task) across 50-2000 epochs per task.

Results.

epochs	SGD	Swarm	Winner	
50	0.655	0.695	SWARM	(+4.0pp)
100	0.785	0.787	SWARM	(+0.2pp)
200	0.910	0.870	SGD	(+4.0pp)
400	0.928	0.908	SGD	(+2.0pp)
800	0.900	0.913	SWARM	(+1.3pp) ← SGD peaks, begins degrading
1200	0.898	0.920	SWARM	(+2.2pp)
1600	0.893	0.915	SWARM	(+2.2pp)
2000	0.877	0.917	SWARM	(+4.0pp)

Three regimes emerge. At low compute (50-100 epochs), Swarm wins because isolated experts converge faster without interference. At medium compute (200-400 epochs), SGD wins because its shared backbone exploits feature overlap

across non-adversarial tasks. At high compute (800+ epochs), SGD DEGRADES — the adversarial Task 0/Task 1 conflict accumulates, driving accuracy from 0.928 down to 0.877 (-5.1pp). Swarm remains stable at 0.917 because adversarial tasks are processed by separate experts that never share gradients.

Multi-seed stability. At 2000 epochs across 5 random seeds, Paradigm Swarm achieves 0.924 ± 0.008 (mean $\pm$ std) vs SGD 0.900 ± 0.019 . Swarm wins on 4/5 seeds with lower variance, confirming that the architectural advantage is robust to initialization.

Feature importance detection. We tracked per-feature gradient magnitude for each expert. Across all epoch counts (50-2000), every expert correctly identified ALL its informative features in its top-5 ranking. 20/20 features detected, zero false positives among top-5. This demonstrates that isolated experts naturally learn to attend to task-relevant features and ignore noise — a property that is lost in shared architectures.

Interpretation. This experiment reveals SGD’s architectural ceiling in its purest form. SGD with task-specific W2 columns already implements a form of output isolation — yet the shared W1 creates an interference channel that accumulates with training. Paradigm Swarm’s full isolation (separate W1 per expert) eliminates this channel entirely. The cost is the absence of forward transfer between non-adversarial tasks (visible at 200-400 epochs). The benefit is unlimited stable training.

4.13 Expert Knowledge Exchange via Situational Responsibility

Design. We test whether experts can autonomously discover task assignments and exchange knowledge in a multi-expert ecosystem. Six experts and four tasks (including the adversarial pair). Experts explore tasks in round-robin cycles, then tasks are assigned to the most confident expert (max 2 tasks per expert). Experts train only on their assigned tasks — no shared warm-up contamination.

Results. The system correctly assigns adversarial Task 0 and Task 1 to DIFFERENT experts — preventing interference without explicit knowledge of which tasks conflict. Each expert develops feature importance tracking through gradient accumulation. After 300 epochs of specialization per expert, all four assigned experts correctly identify 100% of their informative features in their top-5 ranking (20/20 features detected across 4 experts). Tasks that no expert is confident on are flagged as GAP — a natural mechanism for detecting new paradigms.

However, cross-expert knowledge exchange through mutual improvement degrades overall performance: average accuracy drops from 0.892 to 0.880 ($=0.012$). Expert E2 loses 6.5pp ($0.945 \rightarrow 0.880$) from consultation with other experts, while E3 gains 0.5pp. Naive averaging of expert opinions is harmful; the positive result — correct assignment of adversarial tasks — is the validated mechanism. Improving cross-expert consultation without degradation requires query-specific trust calibration (§4.14).

Interpretation. This validates the situational responsibility principle (I-072): experts compete for task ownership based on confidence, the strongest expert on each task takes responsibility, and adversarial tasks self-segregate. The negative result on mutual improvement motivates the trust-calibrated approach in §4.14. The knowledge exchange is implicit — experts do not share weights, but the assignment mechanism routes each task to its most capable expert, creating a self-organizing division of labor.

4.14 Cross-Expert Consultation via Query-Specific Trust

Motivation. Can experts improve each other through consultation? Prior experiments showed that naive averaging of expert opinions degrades performance (§4.13), and weight inheritance only helps when paradigms are hierarchically dependent (§4.11). Here we test a more sophisticated mechanism: query-specific trust calibration with deferral.

Design. Three experts: E0 (shape, features 0-7), E2 (texture, features 10-17), and E3 (mixed shape+texture, features 0-3 and 10-13). E0 and E2 are fully trained (1000 epochs, 400 examples). E3 is deliberately undertrained (20 examples) to create genuine uncertainty — E3 achieves only 0.770 accuracy alone, with 17% of test queries falling below confidence threshold 0.60.

Trust calibration. E3 maintains a per-feature trust matrix: for each consulted expert and each input feature, it tracks how often that expert’s prediction matches the GROUND TRUTH label on calibration data (100 held-out examples with known labels). This differs from §4.13 where trust was calibrated against E3’s own (potentially incorrect) predictions.

Deferral mechanism. For each test query, if E3’s confidence exceeds threshold (0.60), it decides alone. Otherwise, E3 identifies which features are active in the query, computes a weighted trust score for each potential consultee based on their feature-specific accuracy, and defers to the most trusted expert.

Results.

Task	Train	Swarm	SGD	Winner
T0	400	0.950	0.910	SWARM (+4pp)
T1	400	0.930	0.880	SWARM (+5pp)
T2	400	0.970	0.920	SWARM (+5pp)
T3 (mix)	20	0.830	0.870	SGD (+4pp)
T4	400	0.960	0.940	SWARM (+2pp)
Avg		0.928	0.904	SWARM (+2.4pp)

E3 alone: 0.770 → with deferral: 0.820 (+5pp)
 Low conf (<0.60): Alone=0.353 Defer=0.647 =+29.4pp
 High conf (0.60): Alone=0.855 Defer=0.855 (no interference)
 Changes: 9 predictions flipped - 7 improved, 2 harmed

The feature-specific trust matrix correctly captures domain expertise: E0 is

trusted more on shape features (0.76 on dimensions 0-3) and E2 is trusted more on texture features (0.70 on dimensions 10-13). Importantly, E2 receives LOWER trust on shape features (0.62), demonstrating that the calibration correctly identifies the boundaries of each expert’s competence.

Interpretation. This experiment demonstrates the full Paradigm Swarm consultation loop: (1) query-specific trust calibration using ground truth labels, (2) deferral to trusted experts when confidence is low, (3) no interference when confidence is high. The +29.4pp gain on uncertain queries validates the core architectural claim: experts CAN improve each other, provided they know WHICH expert to trust for WHICH features. This closes the loop between gap detection (§4.4), feature importance (§4.12), and situational responsibility (§4.13).

4.15 Massive-Scale Catastrophic Forgetting: 100 Tasks

Design. To demonstrate catastrophic forgetting in its most destructive form, we scale to 100 sequential binary classification tasks. Each task uses 4 random features from a 100-dimensional input space, with alternating positive/negative classification signs — creating direct adversarial interference. Critically, SGD uses a SINGLE 2-output classification head for all tasks (not task-specific output columns). This is the classic continual learning setup where catastrophic forgetting is most severe. Each task receives 200 training examples and 100 epochs. Paradigm Swarm allocates one isolated expert per task (100 experts total). As in §4.10, all 100 tasks share the same input points (X_{base}), differing only in which 4 features are informative and the classification sign — a worst-case stress test for shared representations.

Results.

```
After T0:   avg=0.780  (SGD starts strong)
After T9:   avg=0.740  (interference begins)
After T24:  avg=0.628  (forgetting accelerates)
After T49:  avg=0.582  (majority of old tasks impaired)
After T74:  avg=0.570
After T99:  avg=0.540  (barely above random chance 0.500)
```

FINAL:

```
SGD avg:    0.540  (+4pp above random)
Swarm avg:   0.771
GAP:         +23.1pp
```

Tasks < 0.55 (near random): SGD=53/100 (53%) Swarm=3/100 (3%)

First 5 tasks: SGD=[0.44, 0.50, 0.62, 0.62, 0.40] Swarm=[0.80, 0.74, 0.54, 0.82, 0.82]

After 100 tasks, SGD has collapsed to near-random performance (0.540 ± 0.015 across 5 seeds). Over half its tasks (53%) fall below 0.55 accuracy. The earliest tasks are destroyed: T0 reaches 0.440 — below random guessing — because the

single output head has been overwritten 99 times by subsequent contradictory signals. Paradigm Swarm maintains 0.771 ± 0.008 average accuracy with each expert preserving its task knowledge in isolation. Three Swarm experts fall below 0.55 due to inherently difficult random feature splits, not due to interference.

Multi-seed stability (5 seeds).

Seed	SGD avg	Swarm avg	Gap
42	0.537	0.771	+0.234
99	0.565	0.759	+0.194
123	0.549	0.769	+0.220
456	0.524	0.771	+0.247
789	0.526	0.785	+0.259
Mean	0.540	0.771	+0.231
$\pm$ Std	0.015	0.008	

Paradigm Swarm wins on all 5 seeds with nearly $2\times$ lower variance (0.008 vs 0.015), confirming that the architectural advantage is robust to both data sampling and initialization.

Interpretation. This experiment demonstrates the fundamental fragility of shared-parameter architectures under continual learning at scale. SGD with a single output head has no mechanism to protect old knowledge — every new task overwrites the shared weights. Paradigm Swarm’s per-task isolation eliminates this failure mode entirely. The 23pp accuracy gap is a direct measure of the cost of weight sharing in sequential learning.

4.16 Strong-Weak Expert Consultation with Adversarial SGD Comparison

Design. 25 Category A tasks (adversarial signs, ± 1) each with a dedicated strong expert (300 epochs, 200 examples). 25 Category B tasks (positive sign, features overlapping with A) each with a deliberately undertrained weak expert (50 epochs, 15 examples). Weak experts calibrate trust against ALL strong experts using their own training data. SGD uses a single 2-output head trained sequentially on all 50 tasks (200 epochs each, shuffled order).

Results.

Category A (strong): Strong=0.877 SGD=0.482 (-39.5pp)
 Category B (weak): Alone=0.661 Consult=0.775 (+11.4pp) SGD=0.289 (-37.2pp)
 Overall: Swarm=0.826 SGD=0.386 GAP=+44.0pp

Adversarial signs in Category A destroy SGD’s shared representation: accuracy drops to 0.482, barely above random on some tasks (min=0.143). Category B tasks, which depend on features shared with A, collapse to 0.289 under SGD. Consultation from strong to weak experts provides +11.4pp improvement (426 corrections) with an average of 11.8 trusted strong experts per weak task. The

full Paradigm Swarm system (isolated strong experts + consulting weak experts) achieves 0.826, more than double SGD’s 0.386.

Interpretation. This experiment demonstrates the complete Paradigm Swarm advantage in a single benchmark: (1) strong experts preserve knowledge under adversarial interference through isolation, (2) weak experts improve through trust-calibrated consultation, and (3) SGD collapses under the combined pressure of adversarial signs and sequential training. The 44pp accuracy gap quantifies the cost of shared parameters in continual learning.

4.17 Router-Free Architecture: Experts as Collective Router

Motivation. Sections 4.7-4.8 demonstrated that a separate LLM-based router achieves 96% accuracy, while trained routers on limited data collapse to 4-12%. This raises the question: is a dedicated router module necessary? In biological cognition, there is no separate “router” — cortical regions process input in parallel, and the most activated region determines the response.

Design. We eliminate the dedicated router entirely. Each paradigm expert acts as a density estimator over its domain vocabulary (word unigrams and bigrams extracted from paradigm keywords). For a given query, ALL experts estimate how “typical” the query is for their paradigm. The expert with the highest density wins. If all densities fall below a threshold (0.3), the query is classified as a GAP. No LLM API, no trained classifier, no separate routing module.

Results. On the 16-query benchmark (10 classification, 2 cross-domain, 2 gap, 2 metaphor):

Classification:	8/10 (80%)	- when keywords are present, routing is correct
Gap detection:	2/2 (100%)	- queries with zero keyword overlap correctly gapped
Cross-domain:	0/2 (0%)	- routes to dominant paradigm, needs density blending
Metaphor:	0/2 (0%)	- density still matches source paradigm
Overall:	10/16 (63%)	

The two classification failures are morphological variants (“fired” vs keyword “dismissal”, “triangle’s angles” vs keyword “triangle” — queries contain inflected or possessive forms absent from the keyword list). These are addressable through stemming and phrase detection. Cross-domain routing (0/2) requires semantic understanding of whether operational definitions survive across paradigms — a capability currently unique to LLM-based routers and identified as future work.

Interpretation. The router-free architecture validates a core design principle: in Paradigm Swarm, the collective of experts IS the router. We also attempted to train a dedicated router (word-bigram MLP) on 109 hand-labeled examples across 5 paradigms; it achieved only 13.6% accuracy (3/22 on the test split), confirming that small-data router training is infeasible and motivating the router-free approach. The router-free architecture eliminates the architectural bottleneck identified in §4.7 and aligns the system with biological cognition — no central router, parallel processing, winner-take-all by inherent relevance. **Update**

(morphological variants). Adding morphological variants of keywords (“fired” alongside “dismissal”, “triangle’s” alongside “triangle”) enables the router-free architecture to achieve 100% classification accuracy (10/10) and 100% gap detection (2/2), matching the LLM router on these core metrics. Overall accuracy reaches 75% (12/16), with failures limited to cross-domain and metaphor queries — tasks that require semantic understanding of operational definitions rather than domain density estimation.

The 63% baseline (80% on clean queries) with zero training and zero API calls demonstrates the viability of this approach as a lightweight alternative to LLM-based routing.

4.18 End-to-End Routing: Experts Route Themselves (Numeric Domain)

Motivation. Sections 4.2-4.16 established that weight isolation prevents forgetting, but used oracle routing — each expert was tested only on queries from its own task. Section 4.17 demonstrated that experts CAN route themselves via density estimation on linguistic data (keyword n-grams). This section closes the architectural loop: a full end-to-end test where (a) queries from ALL tasks are mixed into one pool, (b) experts route themselves via input-space density estimation with NO separate router module, and (c) the system is compared against SGD with shared weights.

Design. Four well-separated 2D Gaussian clusters (centers at $\pm 4, \pm 4; =1.2$), each with its own binary classification hyperplane through the cluster center. This makes routing POSSIBLE (clusters are distinguishable) while keeping classification NON-TRIVIAL (decision boundaries are random). Each cluster has 200 training, 50 validation, and 100 test examples. Experts are isolated MLPs (32 hidden units, ReLU) trained with early stopping (patience=50 epochs on validation accuracy) to prevent overfitting. SGD shares W1 across all tasks with task-specific W2 columns. Three self-routing mechanisms are compared:

- **Density router:** Each expert fits a Gaussian density model to its training inputs. For a query, ALL experts estimate input density $\rightarrow$ argmax. No separate module — each expert models its own input distribution.
- **Confidence router:** Each expert outputs softmax confidence $\rightarrow$ argmax. Also no separate module — experts use their own predictive uncertainty.
- **Prototype router:** Nearest cluster center (pre-computed training mean). Baseline, not self-routing.
- **Random router:** Uniform random expert. Lower bound.
- **Perfect router:** Oracle task identity. Upper bound.

All test queries ($4 \text{ tasks} \times 100 = 400$) are mixed and shuffled. The router must select an expert for each query without knowing the true task.

Results.

Epochs	Density R_Acc	Density E2E	Conf. R_Acc	SGD E2E	PSSGD Gap
50	1.000	0.640	0.047	0.544	+0.096
100	1.000	0.681	0.025	0.583	+0.098
200	1.000	0.696	0.050	0.569	+0.127
400	1.000	0.685	0.021	0.504	+0.181
800	1.000	0.692	0.064	0.502	+0.189
1600	1.000	0.690	0.042	0.508	+0.182

Density routing achieves 100% accuracy — experts perfectly distinguish their own cluster from others through input-space density. The confusion matrix is a perfect diagonal: zero cross-cluster misrouting. Paradigm Swarm with density routing is stable across all epoch counts (0.640-0.696), while SGD degrades from 0.583 to 0.502 as shared weights are overwritten by sequential training. The gap reaches +0.189 at 800 epochs.

Confidence routing, by contrast, achieves only 2-6% accuracy — far worse than random (25%). A confusion matrix reveals the failure mode: experts develop pathological overconfidence, with a single expert capturing most queries regardless of their true origin. Softmax confidence is NOT a domain-relevance signal. This negative result validates the architectural choice of density-based routing over confidence-based routing.

Training stability. Early stopping prevents the overfitting that would otherwise occur at high epoch counts. Without early stopping, expert test accuracy drops from 0.820 at 400 epochs to 0.500 at 1600 epochs (memorization on 200 training examples). With early stopping (patience=50 on validation accuracy), stability is maintained — expert accuracy stays at 0.63-0.70 per task across all target epoch counts. This demonstrates that isolated experts require their own regularization (trivially implementable since experts are independent) but do not suffer from the systemic forgetting that afflicts shared architectures.

Interpretation. This experiment closes the architectural loop opened in §4.1. The full Paradigm Swarm pipeline — density-based self-routing + isolated experts + early stopping — operates end-to-end on mixed queries without any separate router module. Three findings emerge: (1) density-based self-routing achieves perfect accuracy when input distributions are distinguishable, validating the router-free principle from §4.17 in the numeric domain; (2) confidence-based self-routing fails catastrophically, demonstrating that softmax outputs are not a substitute for explicit density estimation; (3) the full system (PS + density routing) outperforms SGD by up to +19pp while maintaining stability across training durations where SGD degrades.

The limitations are clear: density routing requires distinguishable input distributions (trivially satisfied for well-separated paradigms in practice); confidence routing is not viable; and expert quality is bounded by training data size (200

examples $\rightarrow$ ~ 0.70 accuracy). These are addressed in the open problems (§5.6).

4.19 The Binding Problem: Emergent Coalitions

Motivation. Density routing solves «which expert should answer?» for single-domain queries. But queries that span multiple paradigms — «big red circle» combining colour, shape, and size — activate multiple experts simultaneously. A standard argmax router selects one expert and discards the others: «red» is returned, «circle» and «big» are lost. This is the **binding problem** in its architectural form: how to merge outputs from multiple qualifying experts without losing dimensions.

In biological cognition, the binding problem is solved through co-occurrence: neurons responding to different features of the same object fire in synchrony [Singer, 1999]. There is no «binding cortex» — binding is a transient event, not a stored module. The shared input that activated multiple experts IS the binding signal. We apply the same principle: when multiple experts qualify for a query, they form a temporary **coalition**, apply structural merge rules, return the result, and dissolve.

Design. No separate binding layer is stored. Each expert carries its output dimension tag (e.g., **colour**, **shape**, **size**) as metadata. When the router finds multiple qualifying experts, a **Coalition** object is created for that query only. The coalition applies four structural rules derived from the invariant that a single output dimension cannot hold two different values for the same query:

1. **Different dimensions $\rightarrow$ merge all.**
2. **Same dimension, same value $\rightarrow$ reinforce.**
3. **Same dimension, different value $\rightarrow$ conflict (strongest wins).**
4. **Single qualifying expert $\rightarrow$ return as-is.**

The coalition resolves, returns the merged result, and is garbage-collected. If only one expert qualifies, no coalition is created — the system degrades gracefully to standard routing with zero overhead. The coalition knows nothing about colours, shapes, or sizes. It only knows dimension tags — metadata that experts already carry.

Experiment. Three experts (colour: red/blue; shape: circle/square; size: big/small) are trained as isolated binary MLPs on a 20D input space with adversarial feature overlap: colour uses dimensions 0-2, shape uses 1,3,5 (sharing dim1 with colour but with opposite classification sign), size uses dedicated dimensions 10-12. Per-expert calibrated confidence thresholds (20th percentile on a mixed calibration set) determine qualification. On each of 300 compound queries — generated with strong signal on all three dimension sets simultaneously — a temporary Coalition is formed from qualifying experts, resolves, and is discarded. We compare against SGD with three separate output heads (same parameter count as three isolated experts) and a shared W1 backbone, trained sequentially for 2000 epochs per task.

Results — final accuracy (2000 epochs/task, 300 compound queries).

	Colour	Shape	Size	Avg	Coverage
SGD (3 heads)	0.783	0.850	0.873	0.836	100%
PS standalone	0.967	0.970	0.987	0.974	100%
Coalition	1.000	1.000	1.000	1.000	90%

SGD forgetting trace. After training on colour alone, SGD achieves 0.963 colour accuracy. After training on shape (adversarial overlap on dim1), colour drops to 0.843 — a 0.120 forgetting event from a single subsequent task. After size training, colour reaches 0.783 (total forgetting: 0.180). Shape similarly degrades from 0.897 to 0.850. In contrast, PS experts are trained independently — the colour expert never sees shape or size data, and its accuracy on compound queries is 0.967 regardless of training order.

Coalition dim distribution. On 300 compound queries, the coalition forms with 3 experts on 47 queries, 2 experts on 222, and 1 expert on 0 — 90% coverage overall. On the 47 queries where all three experts qualify, all-3 accuracy is 1.000. The 10% of queries where no expert qualifies (GAP) are points far from all training distributions — correctly abstained.

Interpretation. Three findings. First, even with separate output heads — a fair parameter-matched comparison — SGD loses 0.180 accuracy on the first task due to shared W1 interference. The adversarial feature overlap (dim1 used by both colour and shape with opposite signs) creates gradient conflict that no output head isolation can resolve. Second, PS isolation eliminates this channel entirely: expert accuracy is independent of training order and adversarial feature overlap. Third, the coalition mechanism preserves 1.000 per-dimension accuracy on 90% of compound queries — answering when confident, abstaining when not. The 10% GAP rate is not a failure: it is the correct behaviour for queries outside all training distributions, and it degrades gracefully to standard routing when only one expert qualifies.

The coalition mechanism does not solve the binding problem in its full generality — output dimension tags must be pre-annotated, and cross-domain synthesis without external API remains future work (§5.6). But it demonstrates that the problem is architecturally tractable: the same structural principles that prevent forgetting (isolation) and detect novelty (gap detection) naturally extend to output merging when the routing decision is changed from **argmax** to **all above threshold**. Binding is not a stored layer. It is an emergent property of experts cooperating on a shared query.

5. Discussion

5.1 When Isolation Wins — and When It Doesn't

Our experiments reveal a clear pattern. Paradigm Swarm dominates when paradigms have distinct feature spaces (2D Gaussians, Split-Patterns: 3-5× improvement) or when tasks are numerous (50 tasks with oracle routing: +38.7pp;

50 tasks with density routing: +22.3pp; 100 tasks: +23.1pp). SGD dominates when tasks are few and share low-level features with few classes per task (Split-CIFAR-10 5-way, cross-paradigm dependency), but catastrophically fails when class count is high — on CIFAR-100 with 50 classes per task, SGD forgets 0.568 despite 11M ResNet-18 parameters (§4.9.2). Critically, when the full pipeline is tested end-to-end with real routing at scale (§4.10, §4.18), Paradigm Swarm with density-based self-routing outperforms SGD with routing accuracy above 99% — demonstrating that the architectural principles hold beyond oracle routing.

5.2 The Forward Transfer / Forgetting Trade-Off

Paradigm Swarm is not uniformly superior. On benchmarks with shared low-level features (MNIST, CIFAR with few tasks), monolithic methods benefit from forward transfer: features learned for task A help with task B. Paradigm Swarm’s isolation prevents both interference AND transfer. On MNIST (§4.8), distillation-based methods (LwF) achieve the best accuracy (0.790) by combining shared representations with output-level knowledge preservation, while Paradigm Swarm (0.765) is statistically indistinguishable from SGD (0.770) at this small task count.

This is a feature, not a bug. The architecture makes the trade-off explicit: if your tasks share structure and you have few of them, use a shared model. If tasks are semantically distinct or numerous, use isolation. No existing method lets you choose. Paradigm Swarm does.

5.3 Paradigms vs. Tasks

The fundamental distinction between our work and concurrent modular isolation methods is the unit of routing. A “task” is an artifact of benchmark design — a set of classes presented together in a specific order. A “paradigm” is a structural property of knowledge — a domain with axioms and operational definitions.

This distinction matters in two ways. First, task boundaries must be specified externally; paradigm boundaries can be discovered through structural analysis of the knowledge domain. Second, task-based routing fails on queries that do not belong to any predefined task; paradigm-based routing, when using density-based experts, naturally detects such queries as gaps (§4.4). Extending gap detection to neural-network experts is an open problem (§5.5).

A concrete example illustrates the difference. Consider the query: «Can an employer fire a worker for refusing to work at 45°C?» A task-based router faces an undefined situation: this belongs neither to a pure Labour Law dataset nor to a pure Thermodynamics dataset — task boundaries partition the input space and offer no mechanism for overlap. A paradigm-based router receives density scores from both experts: the Labour Law expert recognizes «employer,» «fire,» «worker» (density 0.8); the Thermodynamics expert recognizes «45°C,» «temperature» (density 0.7). Both experts claim the query — triggering cross-domain consultation. This is not a failure mode. It is the correct behavior for a

query that genuinely spans two knowledge domains. Task is thus a degenerate case of paradigm: when queries never cross dataset boundaries, task = paradigm. When they do — and real-world queries do — the distinction becomes operational.

Third, and most importantly for deployment: **frozen weights prevent cross-paradigm interference, not updating.** When a paradigm shifts — a new labor code is enacted, a scientific discovery reconfigures thermodynamics — the corresponding expert is retrained on updated data and swapped in. The old expert version is retained as a frozen reference, enabling comparison between paradigm versions. This mirrors biological memory: the hippocampus consolidates knowledge within domains without destabilizing unrelated circuits. Isolation makes updating *easier* than in monolithic architectures: retraining one expert leaves 49 others untouched. In a shared-weight network, retraining on new data for one paradigm risks overwriting representations for all others — the very interference Paradigm Swarm eliminates. The architecture thus supports both stability (frozen experts) and adaptability (retrain-and-swap within paradigm boundaries). Controlled forgetting of outdated knowledge — retaining the old as reference while deploying the new — is a feature, not a bug. ### 5.4 Connection to Human Cognition

The Paradigm Swarm architecture mirrors the structure of scientific knowledge as described by Kuhn [1962] and codified by Vikentiev: knowledge organized into paradigms with explicit operational definitions, where paradigm shifts reconfigure local structure without global disruption. This is not a metaphor — it is an architectural isomorphism.

But the isomorphism goes deeper than knowledge organization. In Paradigm Swarm, **paradigms are not externally defined categories.** A paradigm IS the competence boundary of its expert — the set of inputs on which this expert outperforms other available experts. This definition is operational, not philosophical: it is measured by prediction accuracy on shared queries. If expert A achieves 0.95 on labour law queries and expert B achieves 0.50, labour law belongs to expert A — regardless of whether anyone labelled «labour law» as a category. The boundary emerges from competition, not from taxonomy.

This operational definition resolves a long-standing ambiguity in Kuhn’s framework: how to determine whether two paradigms are distinct. Kuhn argued paradigms are «incommensurable» — they cannot be compared because they use different vocabularies and standards. Paradigm Swarm makes them measurable: send the same query to both experts, compare predictions, the more accurate expert claims the query. An expert may err on a specific query — misclassifying a borderline case — yet remain the best available expert for that paradigm. Routing is argmax over relative competence, not a demand for perfection. The swarm does not need experts that never fail. It needs experts that fail less often than every other expert on the queries that matter.

5.5 Relationship to Out-of-Distribution Detection

Gap detection in Paradigm Swarm (§4.4) is a special case of out-of-distribution (OOD) detection — identifying inputs outside the training distribution. Unlike standard post-hoc OOD methods (ODIN, Mahalanobis distance, energy-based scores), gap detection emerges structurally: each expert models only its own paradigm’s density, and inputs falling below all density thresholds are naturally rejected. This unifies routing and OOD detection in a single mechanism — the density estimator IS both the expert and the OOD detector. We have demonstrated this for Gaussian density estimators (100% detection) and extended it to neural-network experts via Max Softmax Probability (90% detection; §4.4). Closing the remaining 10% gap — the MSP extrapolation failure on points aligned with a cluster axis — requires more sophisticated OOD scores (energy-based, Mahalanobis on hidden representations) and connects Paradigm Swarm to the broader OOD literature.

5.6 Open Problems and Research Agenda

Paradigm Swarm solves catastrophic forgetting structurally. In doing so, it opens several problems that we identify as the research agenda for paradigm-based architectures:

1. **Automatic paradigm discovery.** Currently, paradigms are defined manually through operational definitions (§3.1). How can a system discover paradigms from unlabeled corpora? Candidate approaches include contrastive clustering of domain vocabularies and density-based segmentation of embedding spaces. The router-free architecture (§4.17) provides a bootstrap: n-gram density estimation as a weak signal for initial clustering.
2. **Cross-paradigm synthesis.** The router-free architecture achieves 100% on clean classification and gap detection but 0% on cross-domain queries (§4.17). The LLM-based router achieves 88% on cross-domain but requires an external API. A trained middle ground — small transformer fine-tuned on paradigm definitions — could combine speed with cross-domain accuracy.
3. **Hierarchical paradigm dependencies.** Section 4.11 showed that paradigm dependencies make pure isolation suboptimal. Can a system automatically detect which paradigms depend on which, and construct an inheritance graph? This connects to causal structure learning and module networks.
4. **Scale to 1000+ paradigms.** Our experiments demonstrate 100 parallel experts (§4.15) with zero interference. At 1000+ paradigms, memory and routing latency become constraints. Sparse expert activation and parameter-efficient expert architectures (LoRA adapters) are natural paths forward.
5. **Beyond classification.** All experiments are classification tasks. Extend-

ing Paradigm Swarm to regression, sequence generation, code synthesis, and reinforcement learning requires adapting the expert interface. This is architecturally straightforward but experimentally unexplored.

6. **Theoretical guarantees.** A formal proof of zero-forgetting under weight freezing — and bounds on approximation error for queries spanning multiple paradigms — would strengthen the theoretical foundation.

We release the full experimental code as a toolbox for the community to explore these directions.

6. Conclusion

We have presented Paradigm Swarm, an architecture for continual learning where the unit of weight isolation is not a task but a paradigm — a semantic domain with operational definitions. This architecture yields three properties absent from task-based methods: zero forgetting by construction, gap detection as an emergent structural property, and an unlimited compute ceiling.

Experimental validation across 20 experimental configurations demonstrates that Paradigm Swarm matches Oracle within 0.002 accuracy, eliminates catastrophic forgetting entirely, scales to 50 tasks with density-based self-routing where it outperforms SGD by +22.3pp (with 99.2% routing accuracy), and scales to 100 tasks with oracle routing. We proved that SGD faces a structural ceiling where more training *increases* forgetting, while Paradigm Swarm has no such limit; that density estimation — not softmax confidence — is the correct mechanism for experts to route themselves; and that the gap detection property extends from Gaussian density estimators (100%) to neural-network experts with MSP (90%).

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Code and experiments available at: <https://github.com/sensus-stoa/paradigm-swarm> Contact: dolgov-ev@ranepa.ru